Course Summary

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Agenda

- What have we learned in the course?

- What’s happening now (2026)?

What have we learned in the course?

- When model grows from less than 1G to more than 1T,
  - How to train?
  - How to fine-tune?
  - How to inference?

- The field is driven by
  - Model architecture
  - Hardware

Hardware

- GPU
  - Single Instruction Multiple Thread-> Generalized parallel computing
- TPU
  - Systolic array -> Matrix multiplication
- Both require SRAM, HBM

Networking

- GPU networking
  - NVLink + NVSwitch
  - RDMA: tree topology + network flow load balancing + multipath
- TPU infra
  - ICI: 3D torus
  - Optical circuit switching
- Both requires efficient collective communication (e.g., NCCL)

What are the hardware constraints of training?

- GPUs, Inter-GPU bandwidth

- Components inside GPUs
  - SRAM <-> HBM bandwidth
  - HBM sizes

Model architecture

- Transformer
  - QKV Attention -> MHA, GQA, MLA, Top-K Attention
  - FFN -> MoE
- Autoregressive (masked self attention) transformer -> LLM
- Vision Transformer (ViT)
- Diffusion Transformer (DiT)

Training

- How to automatically convert a forward graph to a backward graph?
  - TensorFlow, JAX, Torch

- In parallel training, what additional constraints we put?
  - Given W, nodes store W are the nodes compute dW

- Use GPUs/TPUs efficiently (through hand-written kernels or compilers)
- Balance/overlap the usage of networking and compute to minimize per iteration time

Using GPUs efficiently

- Tiled Matmul, FlashAttention
  - Using shared memory to reduce the HBM I/O
  - Wrap specialization + TMA to overlap data loading and computation
  - FlashAttention2: Online softmax enables split-Q and streaming KV to compute O
- These kernels are pretty hard to develop in CUDA C++
  - Triton: Python DSL
- FP4/FP8 saves both compute and HBM I/O
  - Block scaling: FlashAttention3, DeepGEMM

Using networking efficiently

- Spanning tree vs. Ring

- AllReduce = Reduce Scatter + AllGather

Parallel training

- Vanilla DP -> Torch DDP
  - Overlapping backward computation with AllReduce
- TP
- PP
  - AFAB -> 1F1B, Zero Bubble PP, Chimera, DualPipe
- SP
  - Ring attention
- EP
- How to combine 5D parallelisms + Rematerialization (ZeRO)?

Scaling law

- When you have GPU budget, do you….
- Train with more training data?
- Train larger models?

Fast LLM inference

- Continuous batching: Decode is memory-bound, needs higher batch size
  - Chunked prefill
- KV cache management
  - Paged attention
  - Radix attention
- Disaggregated prefill and decode

Fine-tuning

- Parameter-efficient FT
  - LoRA, QLoRA
- Reinforcement learning
  - Alignment with human value
  - Improved reasoning: math, coding
  - HybridFlow: Scheduling RL workloads on distributed GPUs
- DeepSeek R1: CoT + RL

Other inference techniques

- Speculative decoding

- Sparse attention

Multi-Modality

- Unifying modalities require heterogenous parallelization strategies

- DiT has introduced new opportunities for inferences
  - FastVideo

Driving force for Systems for ML

- New requirements in ML
  - Multi-agent systems with long contexts (e.g., Claude Code, Codex, Gemini CLI)
  - Tool use, browser use, computer use
  - Trustworthiness of agents
  - Video/image/audio generation, world models, robotics

- New hardware trends
  - ASIC
  - Higher radix NVLink/NVSwitch, silicon photonics

Long-context inference

- KV cache requires significant amount of GPU memory
- Mobile devices (e.g., phone, laptop) have limited memory, blocking long-context inference or large model inference
  - Quantization, compression, distillation
- Model architecture: Mixture of Block Attention (MoBA), Linear Attention, DeepSeek Sparse Attention
- Agent memory management

Tool/browser/computer use

- Model context protocol vs. CLI

- Coding agents: Claude Code, Cursor, Codex, Gemini CLI, Gravity
- Browser: OpenAI Atlas

- Robotics, Laboratories

- How to allow AI use tools efficiently and securely?

Agent Security

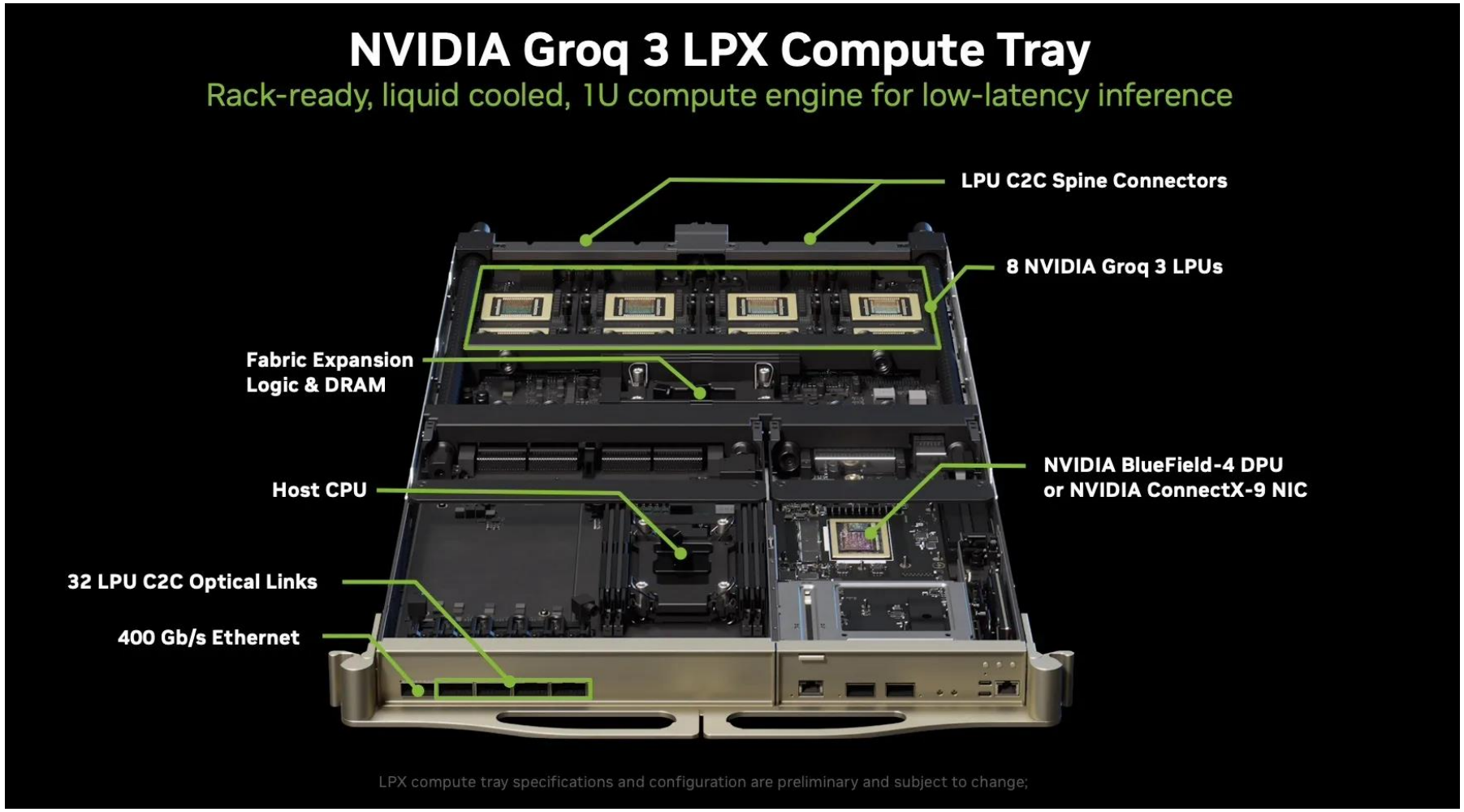
- AI is susceptible to a wide range of attacks
  - Prompt injection attack

- How to enforce security policies so that even agents misbehave, it does not destroy other important components in the systems?
  - Sandboxing for coding agents

New modalities

- Video generation
  - Consistency is hard to achieve
- World models
  - Several paradigms have been proposed. Unclear about the
- Vision language action models (VLA) in robotics

Inference ASIC



Course evaluation

- Please enter course evaluation
  - Help me improve course materials
- My intention is to have the course to be as practical as possible. When you are looking for internship/full time job in the space,
  - Let me know if you feel the course material has helped
  - Let me know if you feel the course material does not help or have some opinions about what I should teach instead to make it more practical
- Good luck on your future endeavors!